Technische Universität Braunschweig



Plasmonic Metasurface Based on Hexagonally Oriented Sn Nanobars

Journal Club: Fundamentals of Nanooptics

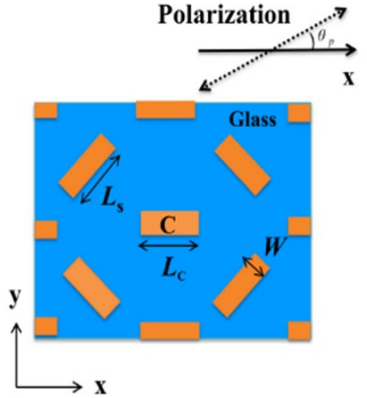
S. Haghi

Prof. Stefanie Kroker

Summer Semester 2025

Abstract

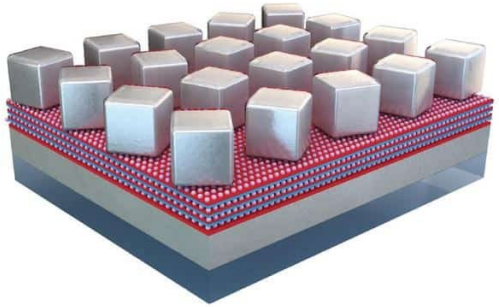
- **Topic:**
  - Plasmonic properties of a Sn metasurface with hexagonal Sn nanobars.
- **Aim:**
  - Resonances, field patterns, and the tunability of resonances.
- **Method:**
  - Designing metasurface.
  - Measuring light transmittance and electric field.
- **Results: (Main part)**
  - **Adjusting the Length of the central nanobar**
  - **Adjusting the Polarization of incident light**
  - **Adjusting the Angle of the central nanobar**
- **Benefit:**
  - A cost-effective way of producing plasmonic material

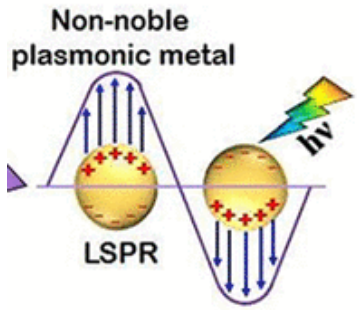


INRODUCTION

Defenitions

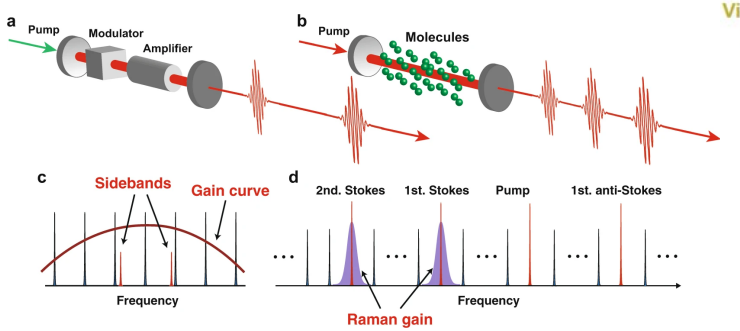
- **Plasmonic metasurfaces**
- **Surface Plasmon (SP)**
- **Surface plasmon resonance (SPR)**

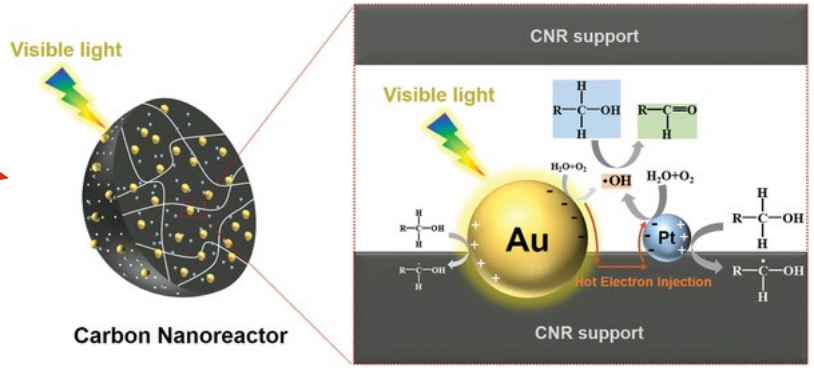


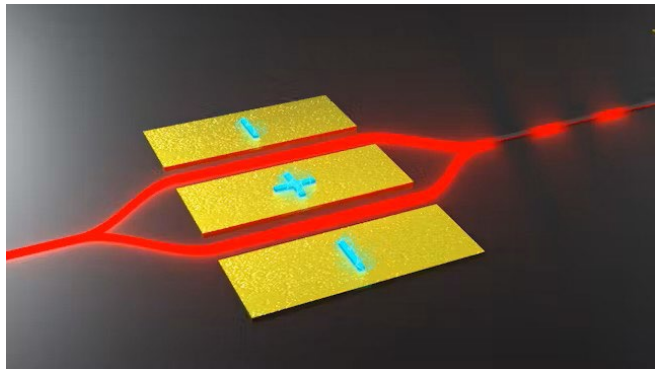


Properties and Applications

- **Exchange of Energy and momentum:**
  - **Quantum sensors**
  - **Neuroengineering**
  - **Nanolasers,**
  - **Nanoantennas**
  - **Biosensing**
- **Controlability, field enhancement:**
  - **Mode-locking lasers**
  - **Advanced devices:**
    - hot-electron nanoreactors
    - nanoscale light sources and modulators

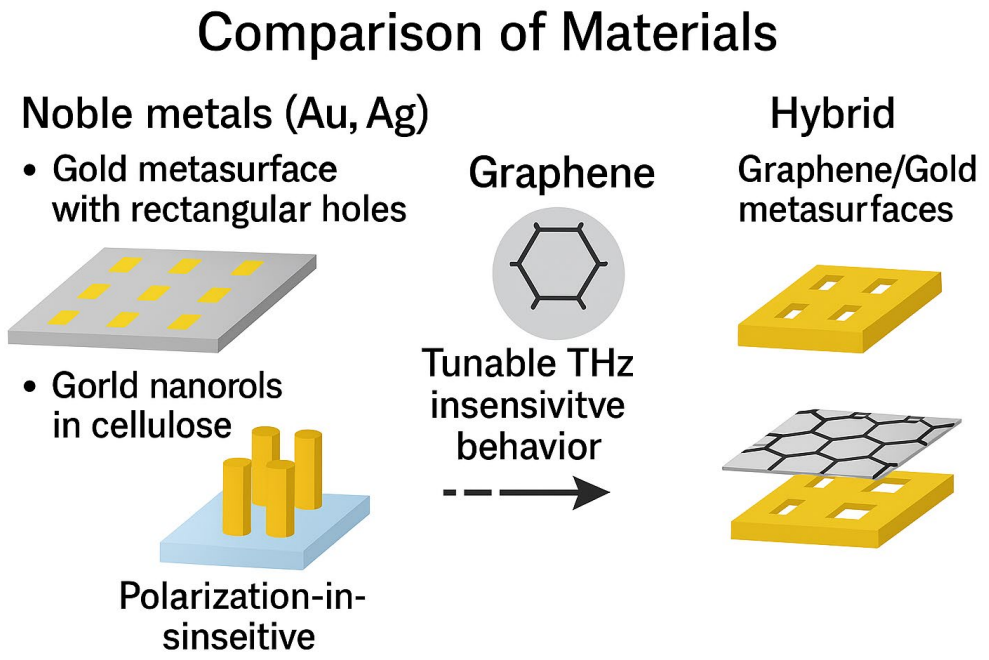






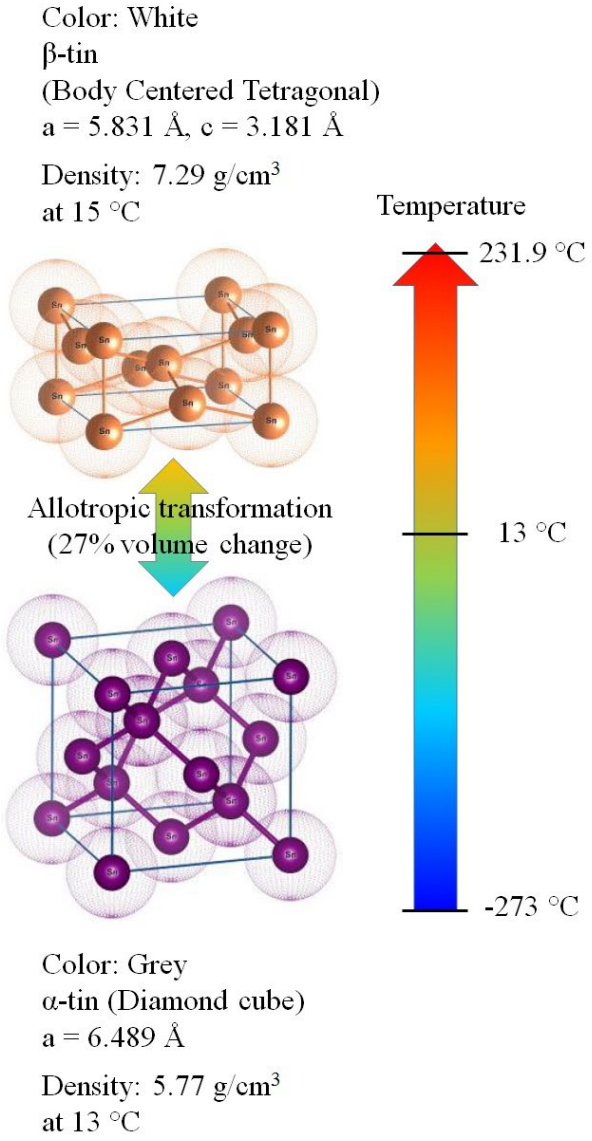
Materials (Others)

- **Noble metals (Au, Ag):**
  - Gold metasurface with rectangular holes
  - Gold nanorods in cellulose
- **Graphene**:
  - Tunable THz response
  - Polarization-insensitive behavior
- **Hybrid**:
  - Graphene/Gold metasurfaces



Materials (Tin)

- **Tin (Sn) in Plasmonics**
- **Advantages over gold**:
  - Low cost
  - Optical flexibility
  - Chemical stability
  - Design flexibility
- **Forms**:
  - α-Sn (semiconductor, cubic)
  - β-Sn (metallic, body centered tetragonal, >13°C)



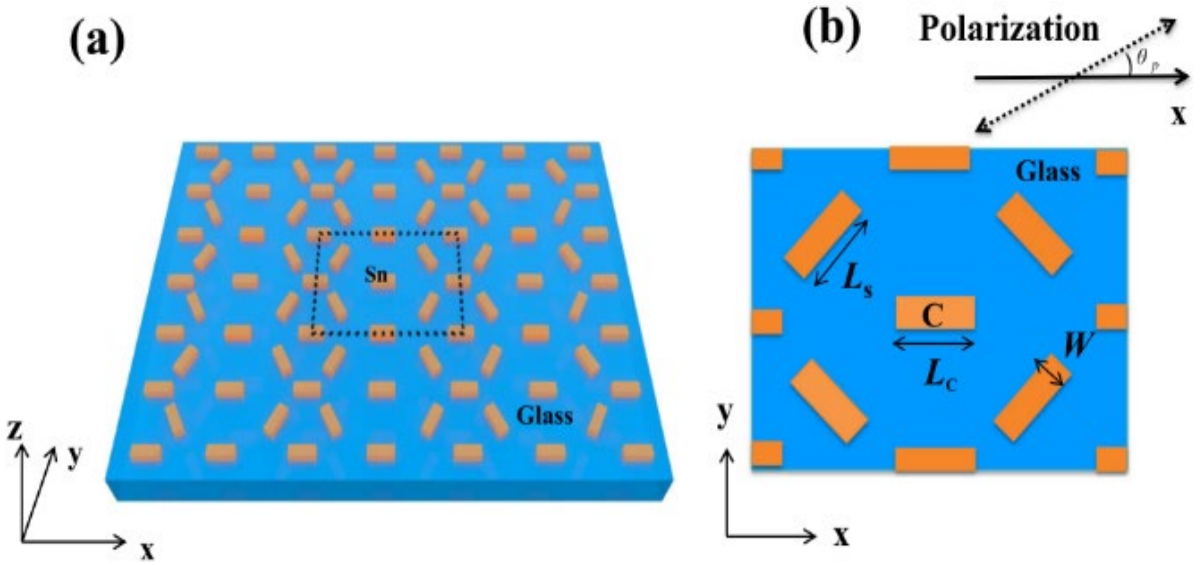
Materials (Tin)

- **Structure**:
  - Central Sn nanobars
  - Surrounding Sn nanobars in hexagonal orientation
- **Goals**:
  - Investigate light transmittance & electric field enhancement
  - Tune resonance wavelength

METHOD

Structure

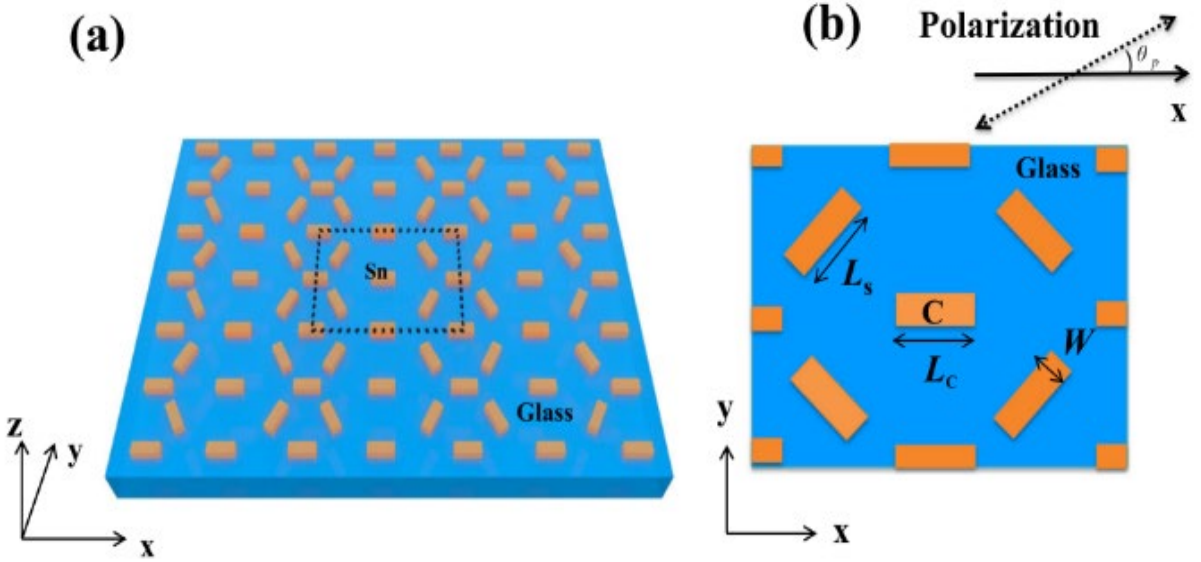
- Metasurface: includes Sn nanobars on a glass substrate
- Structure:
  - Central nanobar labeled **C** with length **Lc**
  - Surrounding nanobars arranged hexagonally with length **Ls = 200 nm**
- All nanobars have:
  - Width **W = 50 nm**
  - Depth **D = 50 nm**



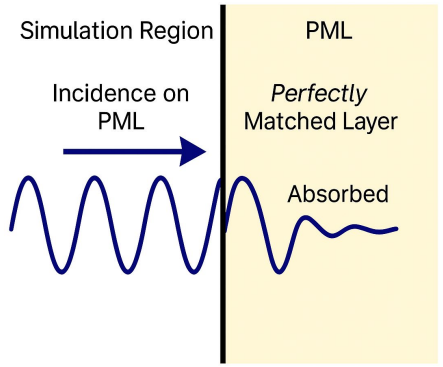
Simulation

- **Setup (FDTD)**
- Method: Finite-Difference Time-Domain (FDTD)

  using Lumerical Solutions
- Advantages: Accurate nanoscale EM modeling; suitable for plasmonic/photonic effect
- Incident light:
  - Plane wave along z-axis
  - Wavelength: **0.4 – 4 µm**
  - Polarization:
    - **θp = 0°** → x-polarized
    - **θp = 90°** → y-polarized



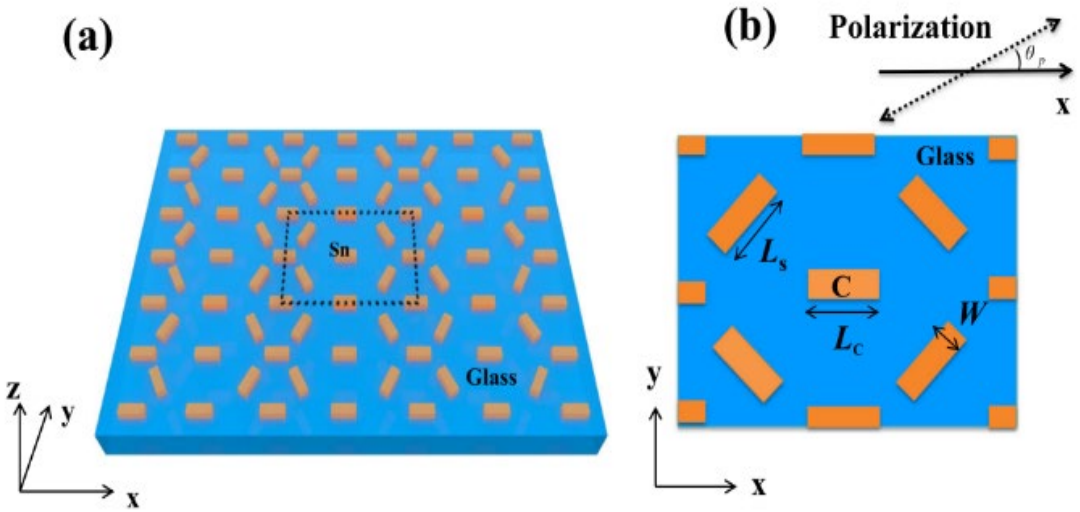
Simulation



- **Boundary & Periodic Conditions**
- Boundary in **z-direction**: Perfectly Matched Layer (PML)
- Structure assumed **infinite in x-y plane**
  - periodic and boundary
- Visualization:

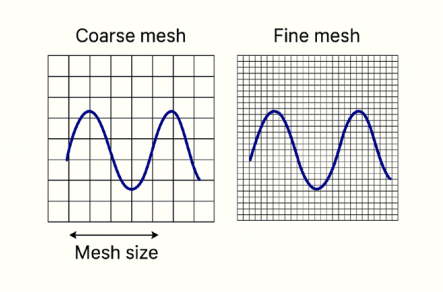
  a shows 3×3 periods

  b focuses on a single period



Simulation

- **Transmittance monitor**: Placed below metastructure (not shown in the figure)
- **Electric field monitor**: Positioned at Sn–glass interface
- **Mesh size**:
  - Less than 1/10 of shortest wavelength

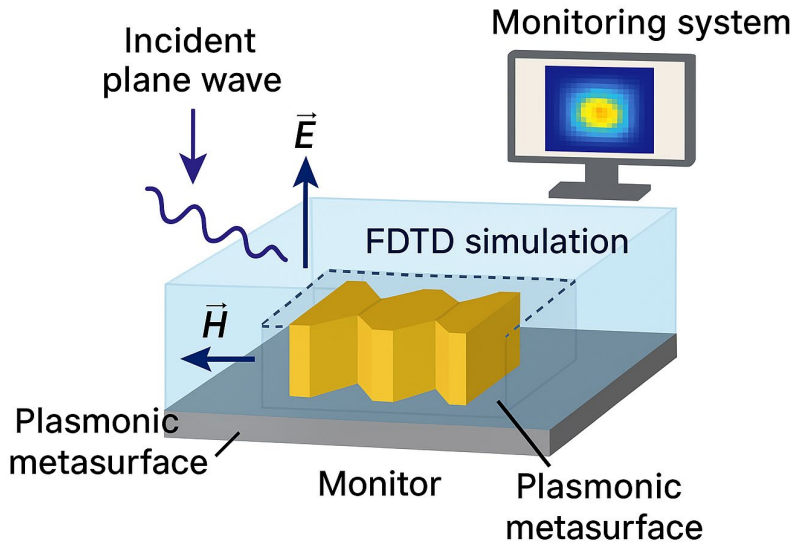


- Sn’s optical constants: Sourced from **Palik’s experimental data**

Monitoring

Output Parameters & Objectives

- Analyzed results:
  - **Light transmittance**
  - **Electric field distribution** at resonance
- Key quantitative parameters:
  - **Resonance wavelength**
  - **Peak field intensity**
- Objective: Understand the plasmonic behavior of the proposed Sn metasurface design



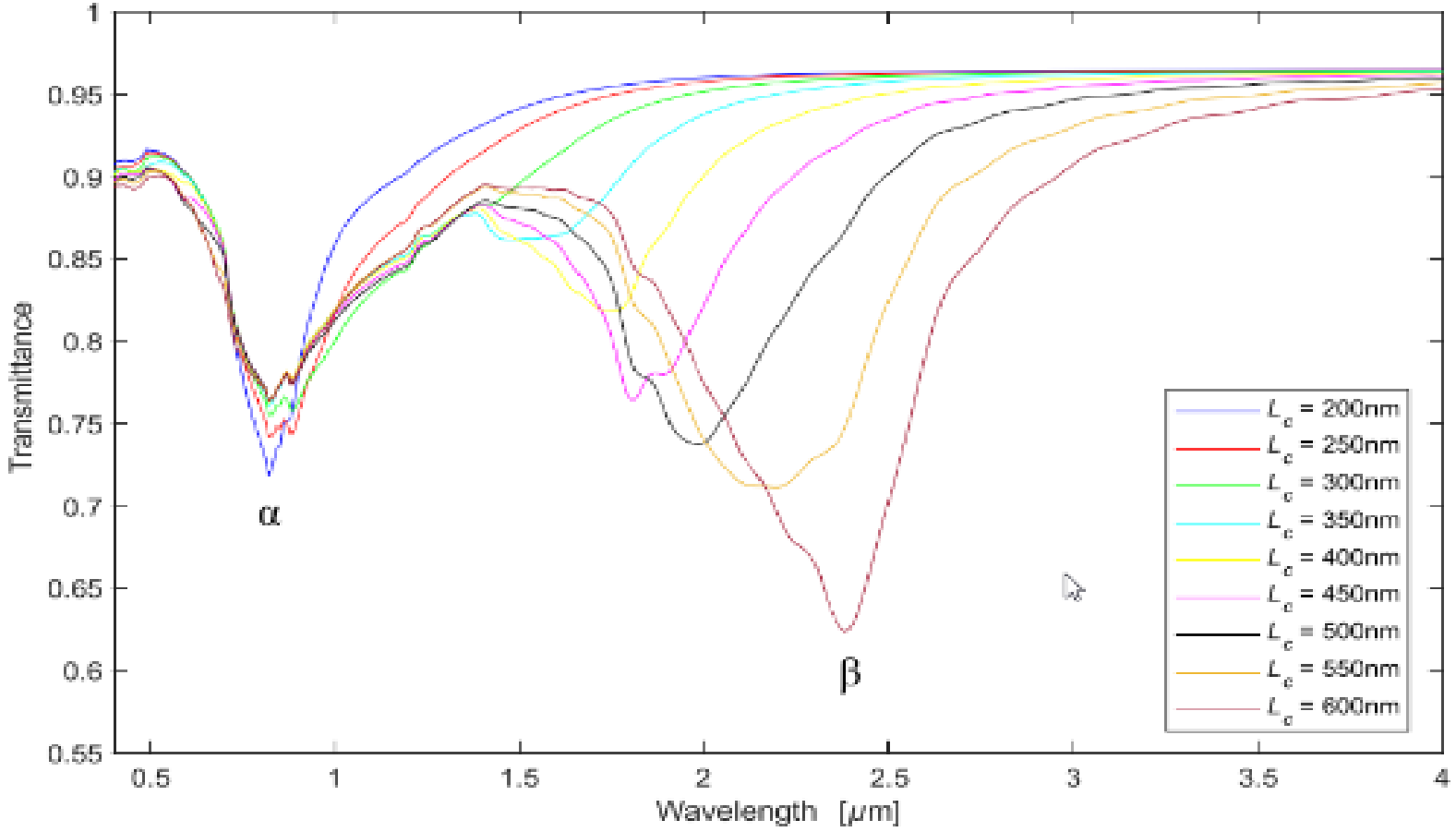
RESULTS

1.Length of the Central Sn Nanobar

- **Simulation Overview**
- Aim: Measuring light transmittance

- Variable: Central bar length **Lc = 200 up to 600 nm**
- Constant: Surrounding bar length **Ls = 200 nm**
- Two resonance peaks observed when **Lc ≥ 600 nm**
  - **α** and **β** → seprated plasmonic resonances

Resonance Behavior



- **Peak α:**
  - Resonance wavelength **remains at 0.82 μm**
  - **Unchanged** peak intensity ≈ 0.236
- When **Lc = 200 nm**, α and β are **indistinguishable**

- **Peak β:**
  - Exhibits **blue shift** as **Lc** decreases
  - Eventually **merges with peak α** when **Lc ≤ 300 nm**

Resonance wavelength & Peak intensity

- Example: At **Lc = 600 nm**:
- **λβ = 2.38 μm**, $M_0 = 0.624 \rightarrow I_0 = 0.376$
- As **Lc decreases from 600 to 400 nm**:
- **λβ ↓ to 1.76 μm, I_0 ↓ to 0.182**
- β peak not derived for **Lc < 400 nm** due to overlap

L_c [nm]	600	550	500	450	400
λ_α [μm]	0.82	0.82	0.82	0.82	0.82
I_0^α	0.236	0.236	0.236	0.236	0.237
λ_β [μm]	2.38	2.16	1.98	1.80	1.76
I_0^β	0.376	0.288	0.263	0.236	0.182

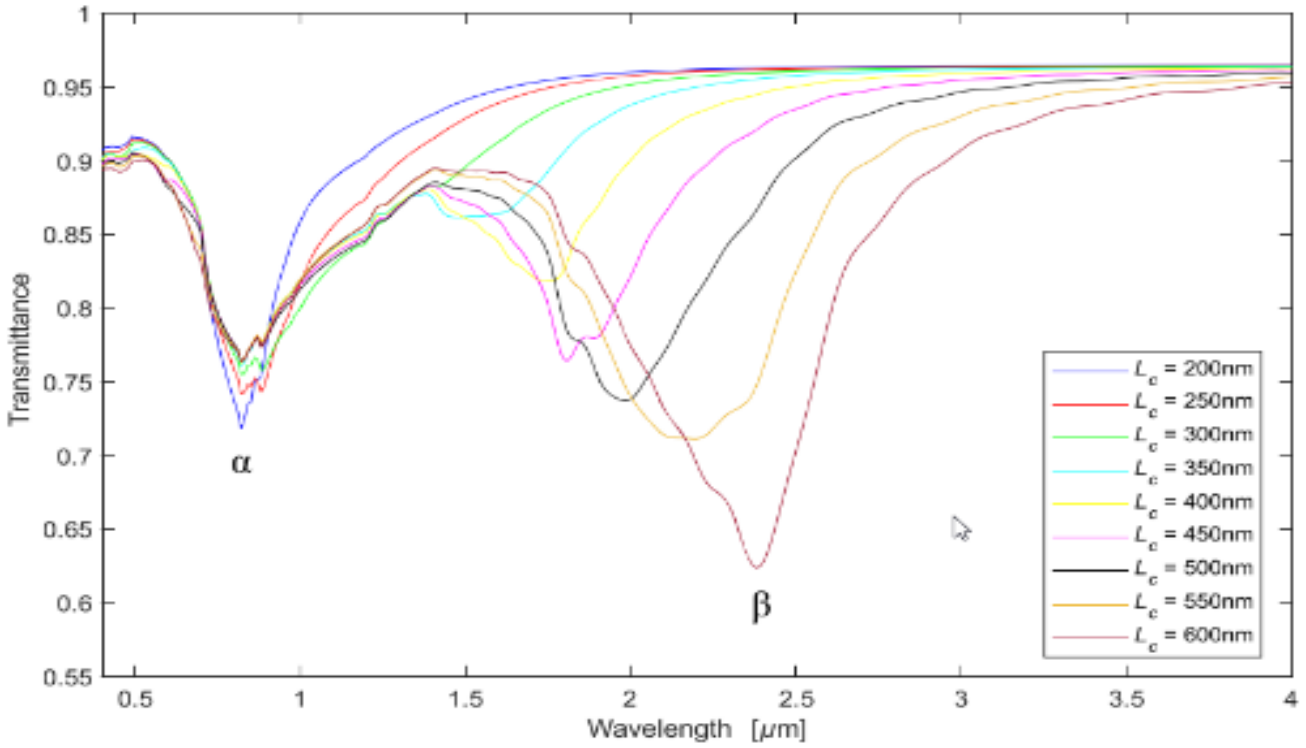


Table 1: The values for the resonance wavelength (λα,λβ), the peak intensity (I α 0 , I β 0), determined from peaks α and β shown in Fig. 2

Electric field distribution

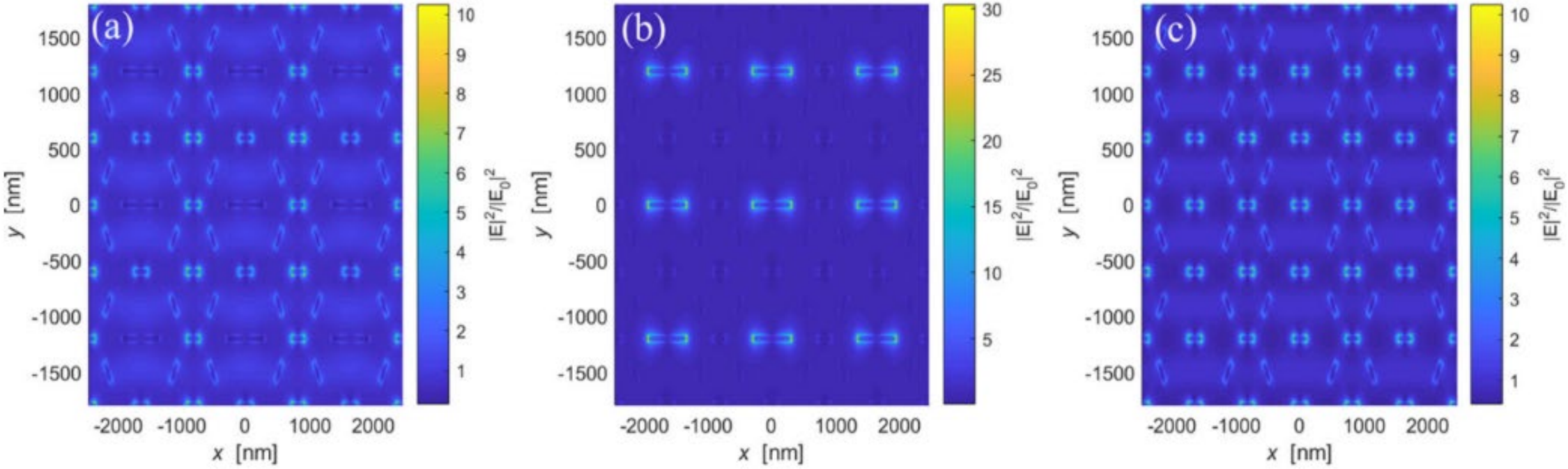
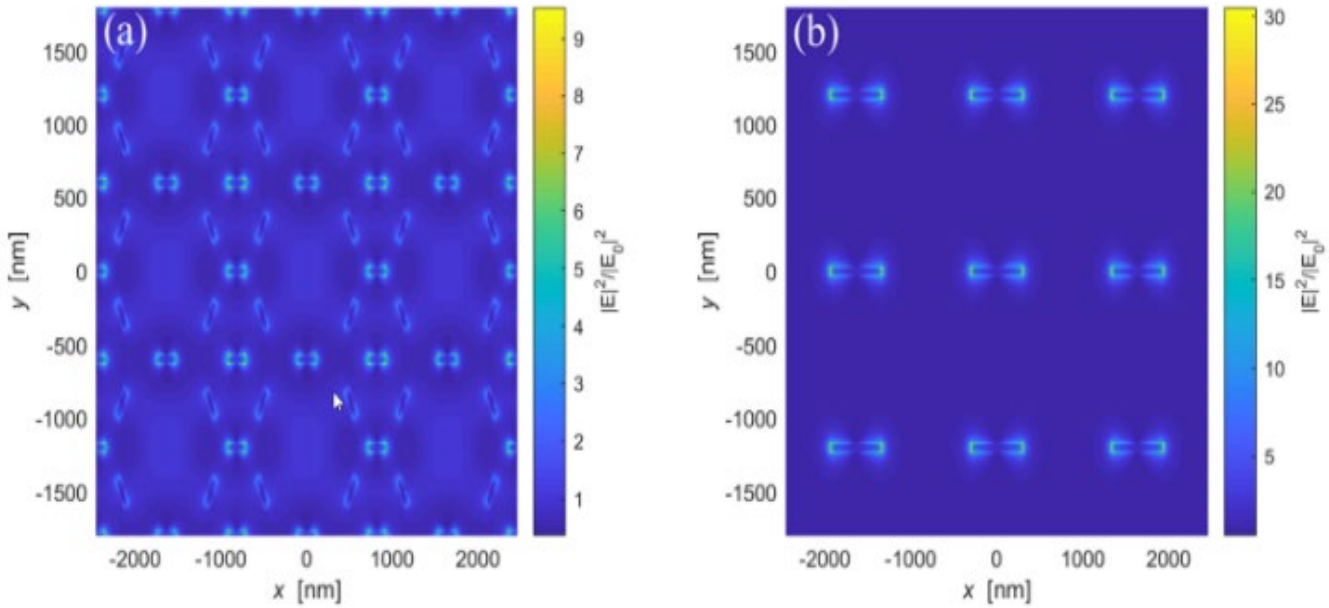


Fig. 3 Electric field distribution in the x-y plane for the cases shown in Fig. 2. The electric fields were determined from the resonance peak according to a peak α of Lc = 600 nm, b peak β of Lc = 600nm, and c peak α of Lc = 200nm.

- **Lc = 600 nm**:
  - **Peak α**: Field localized on **surrounding nanobars**
  - **Peak β**: Field confined to **central nanobar**
- **Lc = 200 nm**:
  - Fields show **mixed pattern**, representing a **summation** of both modes

Electric field distribution



- Simulations with **only central** or **only surrounding** bars at **Lc = 600 nm**
- **Fig. 4a** (surrounding bars only) ≈ **Fig. 3a** (peak α pattern)
- **Fig. 4b** (central bars only) ≈ **Fig. 3b** (peak β pattern)
- Confirms **Fig. 3c** (Lc = 200 nm) shows **combined plasmonic mode**

Fig. 4 Electric field distribution in the x-y plane. The electric fields were determined from the resonance peak according to a peak α, Lc = 600nm, surrounding nanobars only, and b peak β, Lc = 600 nm, central bars only. In the figure, the polarization of the incident light was θp = 0◦



2.Polarization of the Incident Light

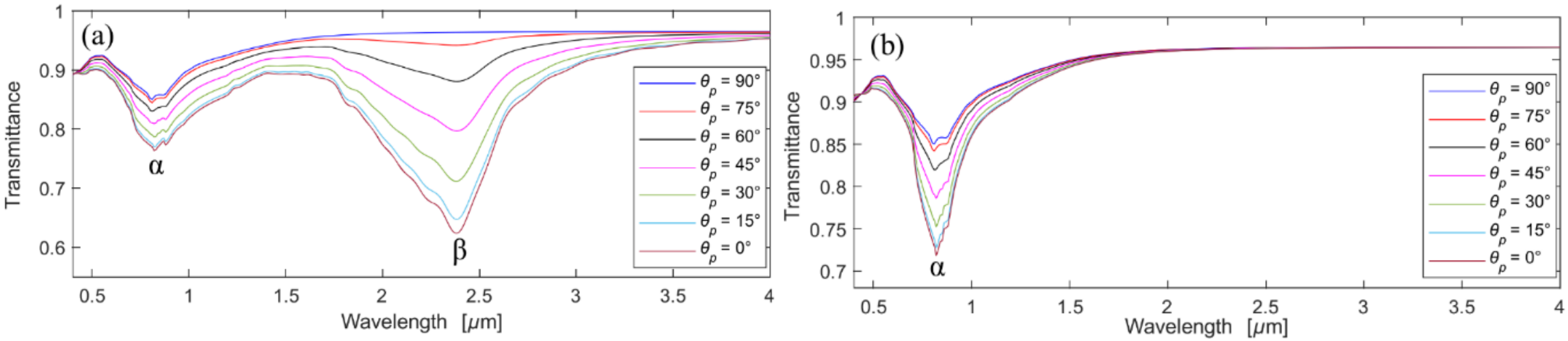


Fig. 5 Wavelength-dependent light transmittance of the structure, as a function of light's polarization. θp was varied from 0 to 90◦.

•**a** (Lc = 600 nm):

- Resonance wavelengths **unchanged**:
  - **β at 2.38 μm, α at 0.82 μm**
- **Intensity of both peaks ↓** as θp increases
- **Peak β vanishes** at θp = 75° or 90°

•**b** (Lc = 200 nm):

- **Single peak** at 0.82 μm remains
- **Peak intensity decreases** with increasing θp

Field Distribution vs. Polarization

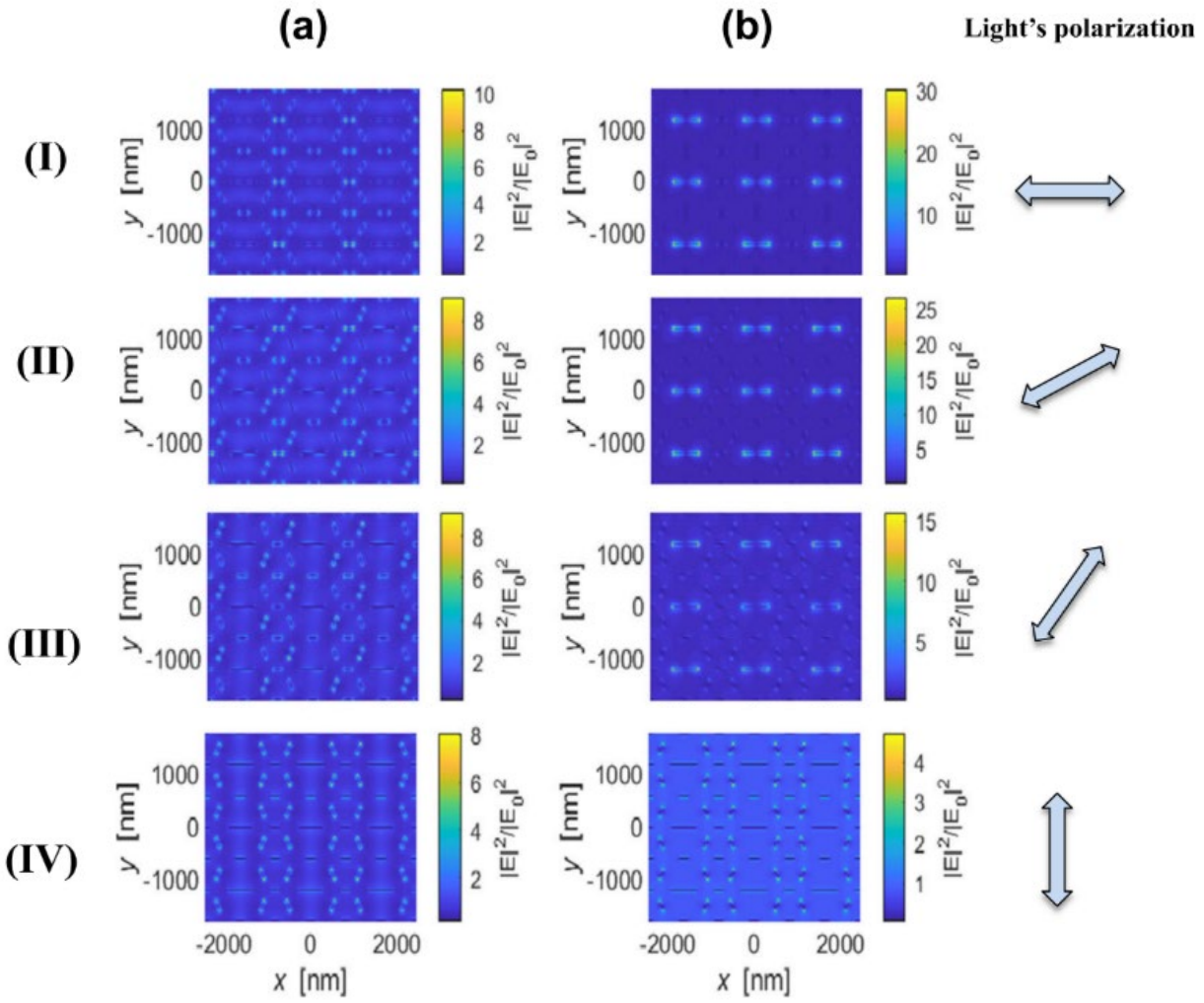


Fig. 6 Electric field distribution in the x-y plane for the cases shown in Fig. 5a (Lc = 600nm). The electric fields were determined from the resonance peak according to (a) peak α , (b) β peaks. In the figure, the light's polarization was θp = (I) 0◦, (II) 30◦, (III) 60◦, and (IV) 90◦.

- **Polarization rotation** from x- to y-direction observed across θp = 0–90°
- **Fig. 6a (peak α)** and **Fig. 6b (peak β)** show:
  - Field pattern **rotates with polarization** (from I → IV)
  - At **θp = 90°**, **field for peak β** aligns along y-axis, **vanishes along x-axis**
  - Explains why **β disappears** in transmittance at θp = 90°

3.Orientation of the Central Sn Nanobar

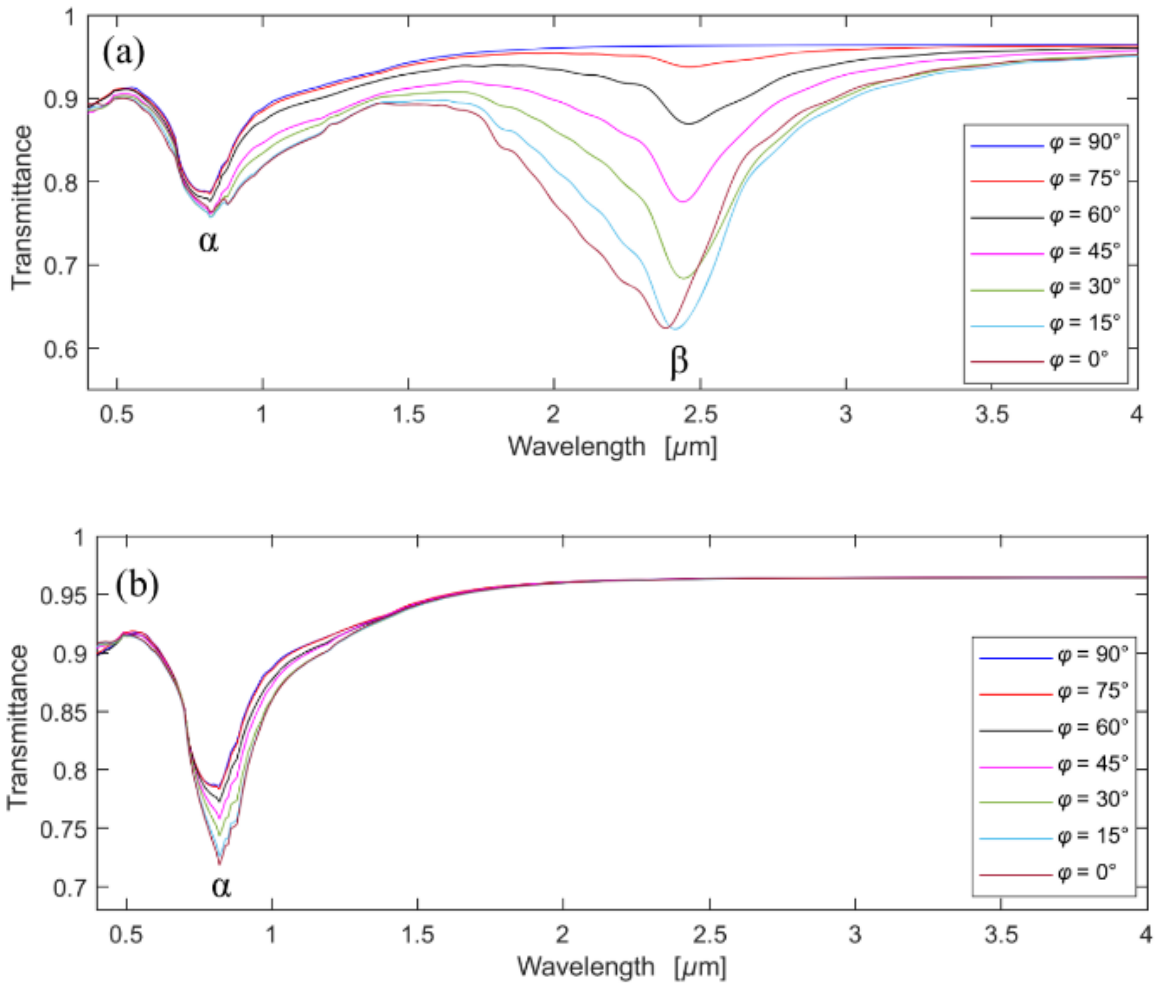


Fig. 7 Wavelength-dependent light transmittance of the structure, as a function of the orientation of the central Sn nanobar. ϕ was varied from 0 to 90◦. a Lc = 600 nm and b Lc = 200 nm. The incident light was x-polarized

- Central Sn nanobar rotated from **ϕ = 0° to 90°** (relative to x-axis)
- **Light kept x-polarized**
- **Peak α**:
  - **Resonance wavelength ~ constant**
  - Appears in both Fig. 7a and 7b
- **Peak β**:
  - **Slight red shift** from **2.36 → 2.39 μm**
  - **Intensity ↓** as ϕ increases
  - Nearly **invisible at ϕ = 75°–90°**

Electric Field Distribution vs. Nanobar Orientation

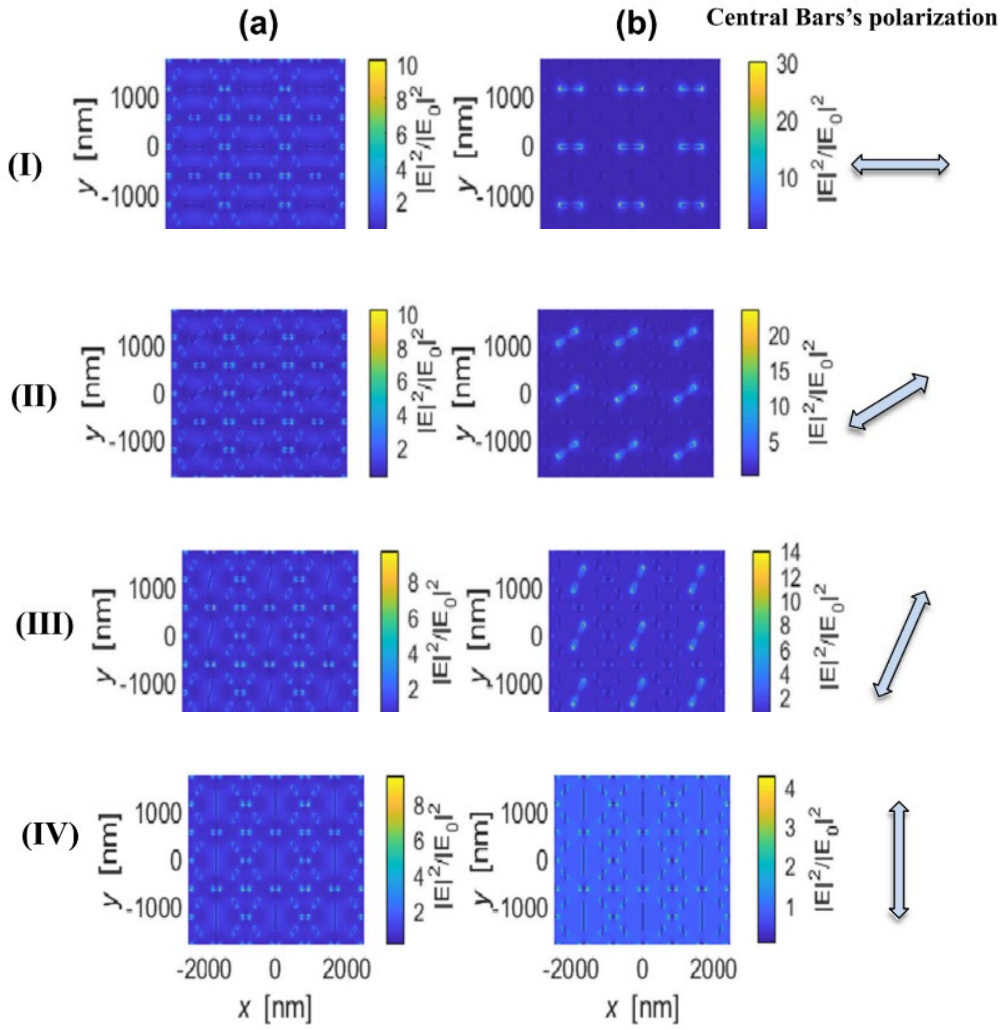


Fig. 8 Electric field distribution in the x-y plane for the cases shown in Fig. 7a (Lc = 600 nm). The electric fields were determined from the resonance peak according to peaks α and β. In the figure, the central Sn nanobar's orientation was ϕ = (I) 0◦, (II) 30◦, (III) 60◦, and (IV) 90◦, as indicated in double arrows

- **Fig. 8a (peak α)**, Lc = 600 nm:
  - Central bar's **field rotates** with ϕ
  - **Surrounding bars unchanged**
- **Fig. 8b (peak β)**:
  - At **ϕ = 90°**, central bar's field **weak**
  - **Surrounding bars' fields become significant**
  - Explains why **peak β fades** at high ϕ values

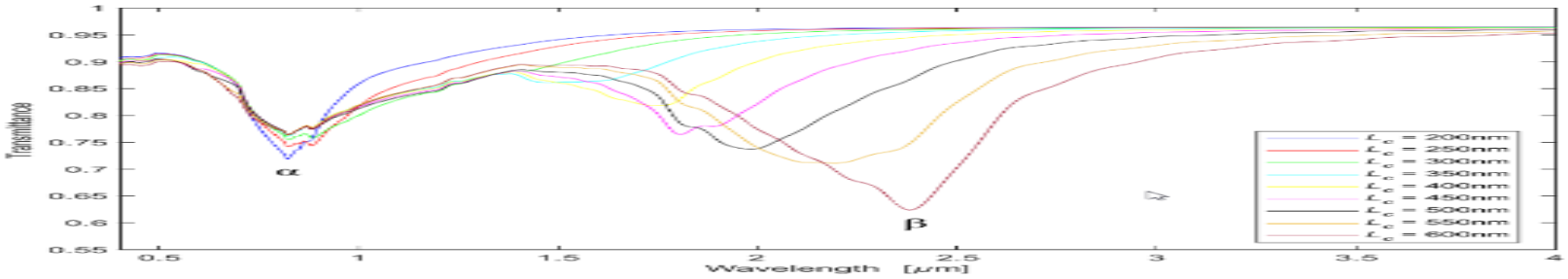
Conclusion

- Proposed a **Sn-based metasurface** with:
  - **Central nanobar + hexagonally arranged surrounding nanobars**
- Simulated **light transmittance (0.4–4 µm)** via **FDTD**
- Identified **two plasmonic resonances**:
  - **λ_1 = 0.82 µm, λ_2 = 2.38 µm**
  - Comparable with other Sn systems:
- **Resonance tunability** via:
  - **Central bar length (Lc)**
  - **Orientation angle (ϕ)**
  - **Polarization angle (θp)**
- **Potential application**: Design guideline for Sn plasmonic devices

Feedback

- **A Valuable New Platform for Nanophotonics**
- **Uses a Cost-Effective → Scalable**
  - **What:** The design uses Tin (Sn) instead of expensive metals like Gold or Silver.
  - **Why it's good:** Tin is “**cheap**” and compatible with standard chip manufacturing (CMOS), making it easier to mass-produce quantum devices.
- **Offers a Highly Tunable Design**
  - **What:** We can precisely control the device's properties by changing the shape and size of the tin nanostructures.
  - **Why it's good:** This allows us to “**tune**” the device to work perfectly with specific quantum systems.

Discussion



Q1: How does increasing the length of the central Sn nanobar affect the field distribution near it?

1. **It enhances plasmonic resonance between central and surrounding nanobars**
2. **It shifts the alpha peak to shorter wavelengths**
3. **It leads to stronger localized electric fields near the central nanobar**
4. **It separates the resonance peaks α and β more clearly**

Discussion

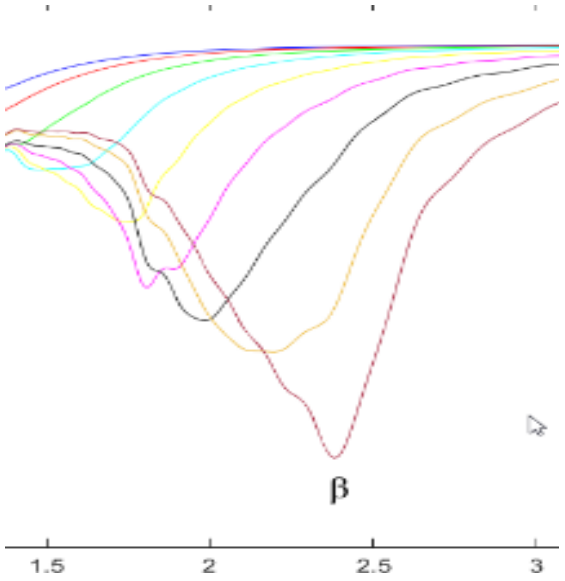
Q2: What distinguishes peak α from peak β based on the simulation results?

1. **Peak α is mainly associated with the surrounding nanobars.**
2. **Peak β is more sensitive to the length of the central nanobar.**
3. **Both peaks appear only under y-polarized light.**

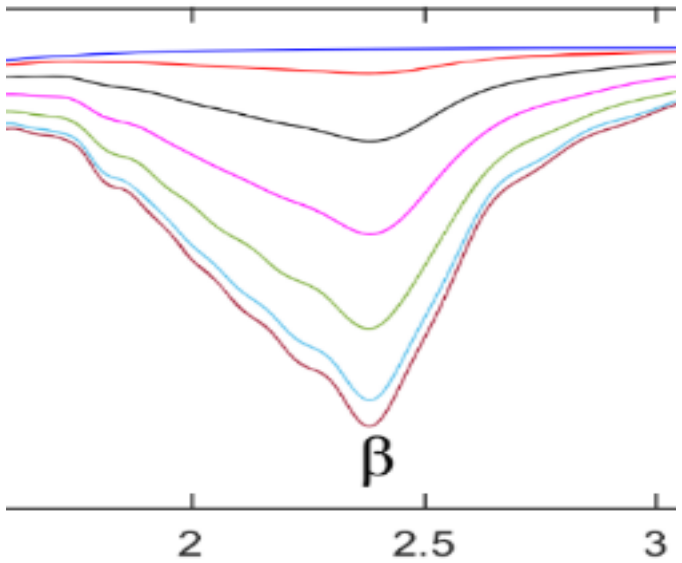
Discussion

- **Q3:When does the redshift in resonance occur?**

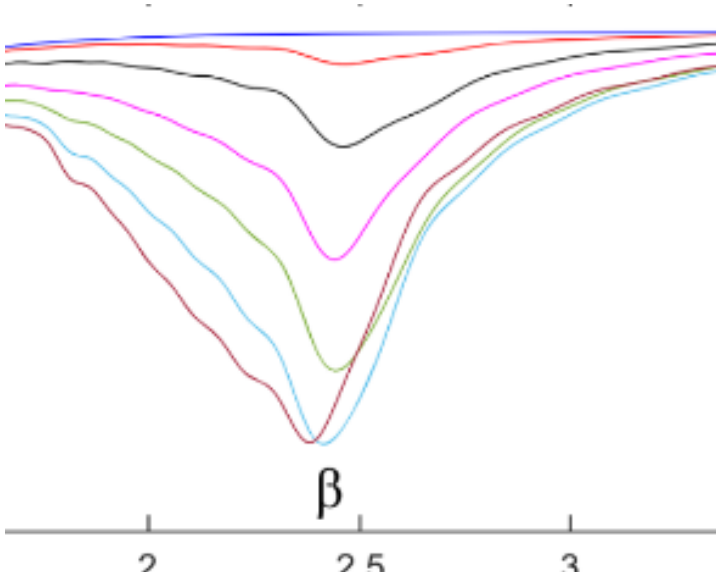
.1



.2



.3



Refrences

- https://doi.org/10.1007/s11468-024-02505-z
- https://www.ansys.com/blog/what-is-fdtd

- Thanks for your attention!